Evangelos Gkiolas

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EDUCATION

University of Patras, Patras, Greece 09/2020 – Present

- Diploma in Electrical and Computer Engineering (5-year Integrated Master; 300 ECTS)
  - Specialization: Signals, Control and Robotics
  - Relevant Coursework: Nonlinear Systems & Control, State-Space Control of Linear Systems, Adaptive Control & Reinforcement Learning, Intelligent Control, Robotic Systems I, Artificial Intelligence I, Linear & Combinatorial Optimization
- Diploma Thesis (in progress): *"Puzzle Assembly via a Robotic Manipulator"*
  - Built a two-stage segmentation pipeline in Python: U^2-Net locates pieces on a downscaled image, then IS-Net re-segments full-resolution crops, fused with the global mask and refined to sub-pixel contours along image gradients (compared U^2-Net, IS-Net and BiRefNet)
  - Detect piece corners via skeleton-based isthmus detection: tabs are cut at their isthmus chord and the convex hull of the remainder yields the four corners
  - Next: edge matching to solve the puzzle and pick-and-place with a robotic manipulator

Politecnico di Milano (PoliMi), Milan, Italy 02/2026 – 07/2026

- Erasmus+ exchange student (MSc level), Electrical Engineering
- Coursework: Advanced and Multivariable Control, Power Generation and Renewables, Smart Grids: Components, Functionalities and Benefits

PROJECTS

Optimal Football Lineup Selection – Binary Integer Programming (Python, Gurobi) 2024

- Course project: modeled starting-XI selection for Olympiacos FC (2023–24) as a BIP with 42 binary player–position variables (27-player squad) and formation/positional constraints
- Built player ratings by combining position-weighted Football Manager 2024 attributes with SofaScore match ratings; solved and compared 5 formations
- Custom sensitivity analysis for integer programs (no shadow prices): rating tolerance intervals per starter and optimal-replacement analysis for absent players
- Optimal 4-3-3 lineup matched 9 of the 11 starters fielded in the 2024 UEFA Conference League final
- Code & report: github.com/Vagkiolas/football-squad-optimization-bip

Katevase To – Desktop YouTube Downloader (Python, Tkinter) 2025

- Desktop GUI app (MP4/MP3) for non-technical users with a thread-safe design (worker thread reports to the GUI via a message queue), self-installing dependencies and a standalone .exe
- Maintained across four releases (v1.2–v1.5); bilingual Greek/English interface
- Code: github.com/Vagkiolas/katevase-to

ACTIVITIES

- Attendee, student conferences of the Hellenic Scientific Society of Electrical & Computer Engineering Students (HSSECE / *SFHMMY*), 2020–2025
- Member, University of Patras Greek Folk Dance Team, 2022–Present

SKILLS & LANGUAGES

Programming: Python, C, MATLAB, Java
Tools & Libraries: Simulink, ROS, Gurobi, OpenCV, NumPy, Git, LaTeX, LTspice, AutoCAD
Languages: Greek (native), English (C2, fluent)